

ENTHUSIAST ADVANCE COURSE

PHASE : MEF-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 21-05-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	1	4	2	1	4	1	2	2	3	3	3	2	1	2	2	2	4	2	4	1	3	3	4	2	2	2	4	2	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	4	3	4	1	3	1	1	4	2	2	1	4	2	4	1	1	2	1	1	2	3	3	3	1	4	1	2	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	4	3	3	2	4	4	1	1	2	4	1	1	1	3	2	1	4	2	3	4	4	2	2	3	4	3	3	1	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	4	3	2	1	4	3	4	3	2	3	2	1	1	1	2	2	4	1	2	1	3	4	4	4	2	1	2	3	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	1	3	3	4	2	3	3	3	2	4	3	4	3	2	4	3	3	3	3	2	2	1	4	3	4	1	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	1	3	3	4	3	1	4	3	3	3	2	2	4	2	4	4	3	3	1	4	1	1	3	2	2	3	2	4	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	3	4	1	1	2	2	2	4	3	1	4	3	2	3	3	3	1	1	4										

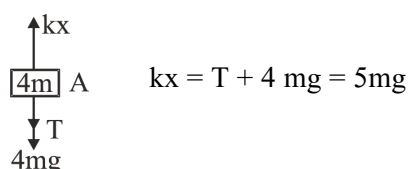
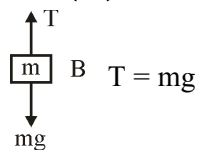
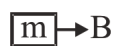
HINT - SHEET**SUBJECT : PHYSICS****SECTION-A**4. **Ans (2)**

$$\text{Potential energy} = -\frac{3GM(M)}{a}$$

Work done = change in PE

$$= -\frac{3GM^2}{2a} - \left(-\frac{3GM^2}{a} \right)$$

$$= \frac{3GM^2}{2a}$$

6. **Ans (4)**After cut the string $T = 0$ 

$$a = g \downarrow$$



$$a = \frac{kx - mg}{4m} = \frac{g}{4} \uparrow$$

7. **Ans (1)**

$$a = g \sin 30^\circ - \mu g \cos 30^\circ$$

$$= g \left(\frac{1}{2} - \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{2} \right) = 0$$

8. **Ans (2)**

$$\text{Kinetic energy} = \frac{p^2}{2m}$$

$\Rightarrow$ if $p \rightarrow 2p$ then kinetic energy $\rightarrow 4$ kinetic energy

So increase kinetic energy $\Rightarrow$ kinetic energy + 3 kinetic energy = 3

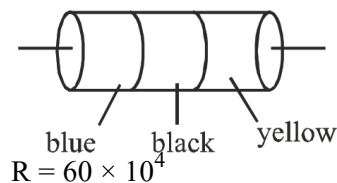
9. **Ans (2)**

Net force acting on a body

$$F_{\text{net}} = (\text{mass}) (\text{acceleration})$$

$$= (2\text{kg}) (5 \text{ m/s}^2)$$

$$F_{\text{net}} = 10 \text{ N}$$

10. **Ans (3)**

$$I = \frac{V}{R} = \frac{30}{60 \times 10^4} = 5 \times 10^{-5}$$

11. **Ans (3)**

Refer Module-04, Pg. # 88, ill 20

12. **Ans (3)**

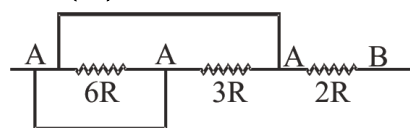
The resistance of the square X is given by

$$R_X = \rho \frac{L}{Lt} = \frac{\rho}{t}$$

where r is the resistivity on the metal.

Similarly,

$$R_Y = \rho \frac{2L}{(2L)t} = \frac{\rho}{t}$$

same as before. Hence, $R_X/R_Y = 1$ 13. **Ans (2)**14. **Ans (1)**

Let I_1 , I_2 and I_3 be the currents flowing through branch Oa, Ob and Oc respectively. If V be the potential of point O, then

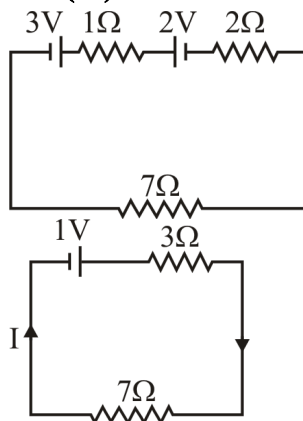
$$I_1 = \frac{6-V}{6}; I_2 = \frac{3-V}{3}; I_3 = \frac{2-V}{2}$$

Using KCL, we get

$$I_1 + I_2 + I_3 = 0$$

$$\text{or } \frac{6-V}{6} + \frac{3-V}{3} + \frac{2-V}{2} = 0$$

$$\text{or } V = 3 \text{ volt}$$

15. **Ans (2)**

The two batteries of emf's 12V and 3V may be replaced by an equivalent battery of emf

$$E = \frac{\frac{12}{6} - \frac{3}{3}}{\frac{1}{6} + \frac{1}{3}} = 2\text{V}$$

and internal resistance

$$R = \frac{(6)(3)}{6+3} = 2\Omega$$

The two batteries in the series may be reduced as

$$E = 3 - 2 = 1 \text{ V}$$

The current through 7Ω resistor is

$$I = \frac{1}{7+3} = 0.1 \text{ A}$$

20. **Ans (4)**

$$\text{By using } \frac{R_1}{R_2} = \frac{(1 + \alpha t_1)}{(1 + \alpha t_2)}$$

$$\Rightarrow \frac{20}{60} = \frac{1 + 20\alpha}{1 + 500\alpha}$$

$$\Rightarrow \alpha = \frac{1}{220}$$

Again by using the same formula for 20Ω and 25Ω

$$\Rightarrow \frac{20}{25} = \frac{\left(1 + \frac{1}{220} \times 20\right)}{\left(1 + \frac{1}{220} \times t\right)}$$

$$\Rightarrow t = 80^\circ\text{C}$$

22. **Ans (3)**

$$(\mu-1)A + (\mu'-1)A' = 0$$

$$(1.5-1) \times 15^\circ + (1.75-1)A' = 0$$

$$0.5 \times 15^\circ = -0.75 \times A'$$

$$A' = -10^\circ$$

23. Ans (3)

$$v = \lambda \times \nu \text{ and } v = \frac{c}{\mu}$$

$$5000 \times 10^{-10} \times 2 \times 10^{14} = \frac{3 \times 10^8}{\mu}$$

$$\Rightarrow \mu = 3$$

24. Ans (4)

$$f = \frac{R}{2(\mu - 1)}$$

$$10 = \frac{R}{2(1.6 - 1)}$$

$$R = 12 \text{ cm}$$

25. Ans (2)

Since aperture of lens reduces so brightness will reduce but there will be no effect on size of image.

26. Ans (2)

Using lens makers formula

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{20} - \frac{1}{-10} \right)$$

$$f = \frac{40}{3} \text{ cm}$$

$$P = \frac{100}{f} = \frac{100}{40/3} = 7.5D$$

27. Ans (2)

$$m = m_o \times m_e$$

$$35 = 7 \times m_e \Rightarrow m_e = 5$$

28. Ans (4)

Snell's law at 2, for TIR

$$\sin \theta = \frac{1}{n} \Rightarrow n = \operatorname{cosec} \theta$$

29. Ans (2)

$$\omega = \frac{n_v - n_R}{n_y - 1} \left\{ n_y = \frac{n_v + n_R}{2} = 1.50 \right\}$$

$$= \frac{0.12}{0.5}$$

$$= \frac{12}{50}$$

$$= \boxed{\omega = \frac{6}{25}}$$

30. Ans (2)

$$u \rightarrow \infty$$

$$v \rightarrow -40 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{-1}{40} - \frac{1}{\infty}$$

$$\boxed{f = -40 \text{ cm}} = -0.4 \text{ m}$$

$$\therefore P = \frac{1}{f \text{ (in m)}}$$

$$= \frac{-1}{(0.4)} D$$

$$\Rightarrow \boxed{P = -2.5D}$$

31. Ans (1)

$$P_{eq.} = P_1 + P_2$$

$$= (4 - 3)D = \boxed{1D}$$

32. Ans (3)

Light bends towards the normal while travelling from rarer to denser medium.

33. Ans (4)

$n_{rel.}$, R_1 & R_2 did not change.

$\therefore P$ remains same.

34. Ans (3)

When a lens is placed in a denser medium, its nature is reversed.

35. Ans (4)

All the given optical instruments use convex lenses.

SECTION-B

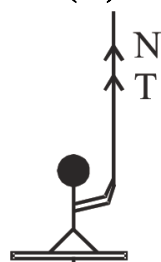
36. Ans (1)

$$\frac{dA}{dt} = \frac{Vr}{2} = 2 \times 10^{16} \quad \uparrow V \propto \frac{1}{r} \downarrow$$

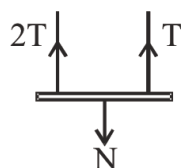
$$V = 20 \times 10^3 \text{ m/sec}$$

$$= 20 \text{ km/sec}$$

38. Ans (1)



80g

For man : $N + T = 80 \text{ g}$... (i)For plank : $3T = N$... (ii)

$$4T = 800$$

$$T = 200 \text{ N}$$

40. Ans (4)

$$\frac{mg}{m(g-a)} = \frac{5}{4} \Rightarrow 4g = 5g - 5a$$

$$5a = g$$

$$a = \frac{g}{5}$$

41. Ans (2)

$$\text{Acceleration} = \frac{10^5}{(10)(1500)} = 6.66$$

$$\text{Force} = (4)(1500)(6.66)$$

$$= 40,000 \text{ N}$$

42. Ans (2)

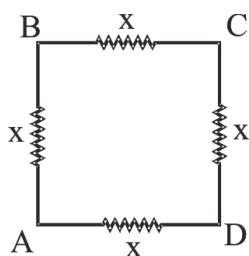
$$V_A = 8 \text{ volt} ; V_B = 2 \text{ volt}$$

$$V_A - V_B = IR$$

$$8 - 2 = I(6) \Rightarrow I = 1 \text{ Amp.}$$

43. Ans (1)

let resistance between A & B is X so CKt can be :-



$$R_{AB} = \frac{3x}{4}$$

$$R_{AC} = x \Rightarrow \text{Ratio} = \frac{3}{4}$$

(i)

46. Ans (4)

$$R = \rho \frac{l}{A} = \rho \frac{l^2}{V} = \rho \frac{l^2}{m} \sigma \quad \left(\because \sigma = \frac{m}{V} \right)$$

$$R_1 : R_2 : R_3 = \frac{l_1^2}{m_1} : \frac{l_2^2}{m_2} : \frac{l_3^2}{m_3} = 25 : \frac{9}{3} : \frac{1}{5}$$

$$= 125 : 15 : 1$$

47. Ans (1)

$$\text{Angular magnification } |m| = \frac{f_0}{f_e} = 5 \Rightarrow f_0 = 5f_e$$

$$\text{distance between objective and eye-piece} = f_0 + f_e$$

$$f_0 + f_e = 36$$

$$5f_e + f_e = 36 \Rightarrow f_e = 6 \text{ cm}$$

$$\text{and } f_0 = 30 \text{ cm}$$

48. Ans (1)

$$\delta = (\mu - 1)A$$

$$\Rightarrow \text{Angle of prism } A = \frac{\delta}{\mu - 1}$$

$$A = \frac{4.8}{1.62 - 1} = \frac{48}{6.2} = 7.74^\circ$$

49. Ans (2)

$$\delta_{\text{prism}} = (n - 1)A$$

$$= (1.5 - 1)4^\circ$$

$$= 2^\circ$$

Now,

$$\delta_{\text{net}} = 90^\circ$$

$$\Rightarrow \delta_{\text{mirror}} + \delta_{\text{prism}} = 90^\circ$$

$$\Rightarrow \delta_{\text{mirror}} = 90^\circ - 2^\circ$$

$$\Rightarrow \delta_{\text{mirror}} = 88^\circ$$

$$\Rightarrow 180^\circ - 2i = 88^\circ$$

$$\Rightarrow 2i = 92^\circ$$

$$\Rightarrow i = 46^\circ$$

$\therefore$ Normal must be rotated by 1° anticlockwise.

50. Ans (1)

$$\text{Above AB} \rightarrow \frac{1.6}{v} - \frac{1}{-2} = \frac{1.6-1}{1} \Rightarrow v = 16\text{m}$$

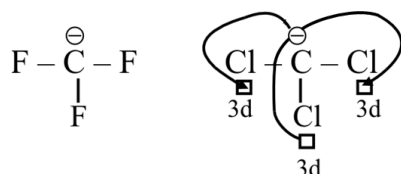
$$\text{Below AB} \rightarrow \frac{2.0}{v} - \frac{1}{-2} = \frac{2-1}{1} \Rightarrow v = 4\text{m}$$

$$\therefore \text{Distance between 2 images} = (16-4)\text{m} = 12\text{m}$$

SUBJECT : CHEMISTRY**SECTION-A**

52. Ans (2)

Conjugate base

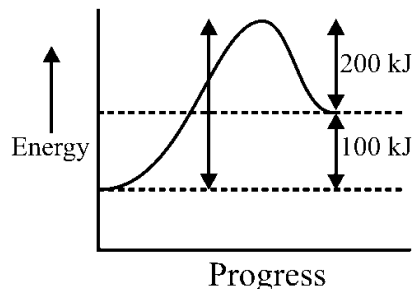
-Ve stable by d orbital resonance in CCl_3^-

64. Ans (3)

$$+\frac{d[\text{HI}]}{dt} = 2 \left(-\frac{d[\text{H}_2]}{dt} \right)$$

$$= 2 \times 10^{-4} \text{ mol L}^{-1} \text{ S}^{-1}$$

69. Ans (1)



73. Ans (1)

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{2} \log_{10} \frac{[\text{Cu}^{+2}]}{[\text{Ag}^{+}]^2}$$

76. Ans (2)

$$\Lambda_{\text{eq}}^{\infty}(\text{BaCl}_2) = \Lambda_{\text{eq}}^{\infty}(\text{Ba}^{+2}) + \Lambda_{\text{eq}}^{\infty}(\text{Cl}^-)$$

$$\Lambda_{\text{eq}}^{\infty}(\text{BaCl}_2) = 127 + 76$$

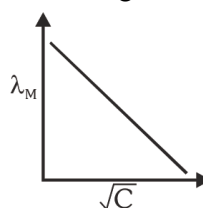
$$\Lambda_{\text{eq}}^{\infty}(\text{BaCl}_2) = 203 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

$$\Lambda_{\text{eq}}^{\infty}(\text{BaCl}_2) = \frac{\Lambda_{\text{m}}^{\infty}(\text{BaCl}_2)}{2}$$

$$\Lambda_{\text{m}}^{\infty}(\text{BaCl}_2) = 2 \times 203 = 406 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

77. Ans (1)

For strong electrolyte



78. Ans (4)

for conc. cell $E_{\text{cell}}^{\circ} = 0$

$$\Delta G^{\circ} = 0$$

and for spontaneous cell reaction

$$E_{\text{cell}} > 0 \text{ and } \Delta G < 0$$

79. Ans (2)

electrolysis of dil. $\text{NaCl}_{(\text{aq})}$ during oxidation and Reduction of water takes place at both electrodesAnode : O_2 Cathode : H_2

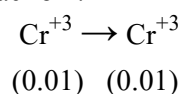
80. Ans (3)

$$G^* = \text{cell constant} = \frac{\ell}{A}$$

81. Ans (4)

At Anode : $\text{Cr(s)} \rightarrow \text{Cr}^{+3}$
(0.01 M)At Cathode : $\text{Cr}^{+3} \rightarrow \text{Cr(s)}$
(0.01 M)

Overall reaction :



$$E_{\text{cell}}^{\circ} = 0$$

$$E_{\text{cell}} = 0$$

82. Ans (4)

order may be +ve, -ve or fraction or zero.
molecularity may be only 1, 2 or 3

83. Ans (2)

$$kt = \ell n \left(\frac{A_0}{A_t} \right)$$

$$\text{or } kt = 2.303 \log \left(\frac{A_0}{A_t} \right)$$

$$t_{1/2} = \frac{0.693}{k}$$

84. **Ans (2)**

E_a , RoR and Rate constant (k) depend on temperature.

85. **Ans (3)**

Unit of A will be unit of k
 for 1st order reaction $k = \text{time}^{-1}$
 it may be sec^{-1} or min^{-1}

SECTION-B94. **Ans (2)**

Due to same electricity

$$\left(\frac{w_1}{E_1} \right)_{\text{Na}} = \left(\frac{w_2}{E_2} \right)_{\text{Al}}$$

$$\frac{4.6}{23} = \frac{w_2}{9}$$

$$w_2 = 1.8 \text{ g}$$

98. **Ans (4)**

moles ratio $\frac{1}{1} : \frac{1}{2} : \frac{1}{3}$
 $6 : 3 : 2$

99. **Ans (3)**

$$\frac{w_1}{w_2} = \frac{E_1}{E_2}$$

$$\frac{46}{w_{\text{Cl}_2}} = \frac{23}{35.5}$$

$$w_{\text{Cl}_2} = 35.5 \times 2 = 71 \text{ g}$$

100. **Ans (2)**

for n^{th} order reaction

$$t_{1/2} \propto (A_0)^{1-n}$$

for 2nd order reaction

$$t_{1/2} \propto (A_0)^{-1} \propto \frac{1}{A_0}$$

SUBJECT : BIOLOGY-I**SECTION-A**101. **Ans (3)**

NCERT XII pg.# 34(E), 36(H)

102. **Ans (2)**

NCERT XII Pg.#38

103. **Ans (1)**

NCERT (XII) Pg. # 35

105. **Ans (1)**

NCERT XII Pg. # 39(E), 41(H)

111. **Ans (1)**

NCERT XIIth Pg # 250,251

112. **Ans (3)**

NCERT Pg # 182

113. **Ans (4)**

NCERT XII Pg. # 273

116. **Ans (2)**

NCERT Pg. # 237

130. **Ans (3)**

NCERT Pg. # 152

134. **Ans (4)**

NCERT Pg. # 154 (E)

SECTION-B138. **Ans (3)**

NCERT Pg. # 220

139. **Ans (3)**

NCERT (XII) Pg. # 251

SUBJECT : BIOLOGY-II**SECTION-A**152. **Ans (1)**

NCERT XII Page # 161

154. **Ans (3)**

NCERT XII Pg. # 72

156. **Ans (3)**

NCERT-XII Pg. # 75

157. **Ans (1)**

NCERT XIIth Pg.#75

164. **Ans (4)**

NCERT (XII) Pg. # 78 (E), 86 (H)

174. **Ans (3)**

NCERT (XII) Pg. # 85

181. **Ans (1)**

NCERT (XIIth) Pg. # 154 (para-8.2.7)

182. **Ans (3)**

NCERT (XIIth) Pg. # 153(E), 165(H)

SECTION-B**186. Ans (2)**

NCERT (XII) Pg # 76-78

189. Ans (4)

NCERT (XII) Pg. # 77

190. Ans (3)

NCERT Pg. # 83

191. Ans (1)

NCERT XII Pg. # 83

193. Ans (3)

NCERT Page # 289

195. Ans (3)

NCERT XII Pg. # 88.

197. Ans (3)

NCERT XII Pg. # 152

200. Ans (4)

NCERT-XII Pg # 90; Para-5.6.2